Milestone 3 Report: Model Fine-Tuning and Optimization

Disease Prediction System

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1 Abstract

This milestone report details the optimization and fine-tuning strategies implemented to improve the core machine learning pipeline of our Disease Prediction System. The primary focus of this phase was to transition from a baseline model to a highly optimized XGBoost classifier capable of accurately differentiating between 727 distinct medical conditions based on 328 binary symptom features. By implementing Bayesian hyperparameter optimization via Optuna and subsequently adopting a leaf-wise growth policy (`lossguide`), we successfully maximized multi-class prediction accuracy while establishing strict computational boundaries to prevent overfitting.

2 Introduction

In the previous milestones, the foundational data processing and baseline model architecture were established. However, dealing with a highly dimensional output space (727 distinct classes) introduces significant challenges, primarily label ambiguity and data sparsity. Analysis of the dataset revealed a theoretical ceiling on Top-1 accuracy of approximately 89.59%, as roughly 17% of the rows share identical symptom patterns with different diseases. Therefore, the objective of Milestone 3 was to push the model's performance as close to these theoretical limits as possible through systematic fine-tuning.

3 Methodology I: Bayesian Optimization with Optuna

To transcend manual parameter guessing, we integrated the Optuna framework to conduct an automated Bayesian hyperparameter search.

3.1 Search Space Definition

The initial search space evaluated standard depth-wise growth configurations. The hyperparameters tuned included:

- **Maximum Depth (`max_depth`):** Bounded between 4 and 8 to prevent memory bottlenecks during the generation of 727 trees per iteration.
- **Learning Rate (`eta`):** Searched on a logarithmic scale (0.05 to 0.2) to find the optimal convergence speed.
- **Regularization (`reg_lambda`, `reg_alpha`):** Tuned to penalize overly complex models and prevent memorization of noise.

3.2 Computational Challenges

During the early Optuna trials, it was observed that unconstrained `max_depth` (e.g., depth 11) caused severe computational latency, taking upwards of 45 minutes per trial even on a T4 GPU. By capping the depth at 8 and raising the minimum learning rate, the search time was significantly optimized, allowing the Bayesian optimizer to iterate quickly and identify patterns without exceeding hardware limitations.

4 Methodology II: Leaf-Wise Growth Strategy

While depth-wise tuning improved some metrics, Top-1 accuracy plateaued. To break this plateau, the architecture was fundamentally shifted to a leaf-wise tree growth policy (`grow_policy="lossguide"`).

4.1 Theoretical Advantage

Unlike depth-wise growth, which expands the tree uniformly level by level, leaf-wise growth expands the tree at the nodes where the loss reduction is the highest. For a dataset with 727 classes and purely binary features, specific symptom combinations carry the majority of the signal. Leaf-wise growth allocates tree capacity much more efficiently by prioritizing these critical splits.

4.2 Final Hyperparameter Configuration

The final tuned configuration utilized for the model (`xgb_tuned_lossguide.json`) is as follows:

- **Growth Policy:** `lossguide` (Uncapped depth, bounded by leaves).
- **Maximum Leaves (`max_leaves`):** 256. This acts as the primary constraint on tree size.
- **Learning Rate (`eta`):** 0.06 (A slower rate for stable convergence over 2000 boosting rounds).
- **Minimum Child Weight (`min_child_weight`):** Increased to 4 to prevent the creation of over-specific, overfit leaves.
- **L2 Regularization (`reg_lambda`):** Increased to 2.2 to provide extra guardrails against the aggressive nature of leaf-wise growth.
- **Max Delta Step:** 1, aiding in stability for highly imbalanced class updates.

5 Results and Performance Metrics

The model was trained on 151,680 samples and validated against an unseen test set of 37,921 samples. The shift to leaf-wise growth significantly enhanced the model's ranking ability, specifically pushing the Top-3 and Top-5 accuracies much closer to their respective theoretical ceilings.

5.1 Accuracy Evaluation

Below is the detailed breakdown of the final tuned model's performance compared against the absolute theoretical ceilings dictated by dataset ambiguity.

Metric	**Achieved Score**	**Percentage**
Top-1 Accuracy	0.7939	79.39%
Top-3 Accuracy	0.9218	92.18%
Top-5 Accuracy	0.9518	95.18
Weighted F1-Score	0.7878	78.78%

Table 1: Final Fine-Tuned Model Accuracy Metrics (Leaf-Wise Growth)

6 Conclusion

Milestone 3 successfully established a highly robust, deployment-ready machine learning model. By strategically applying Bayesian optimization and transitioning to a `lossguide` growth policy, we maximized the model's ability to navigate the complex 328-feature symptom space. The final Top-5 accuracy of 95.18% ensures that when integrated with the backend and frontend systems in the upcoming milestone, the system will provide highly reliable differential diagnosis rankings for end-users.